

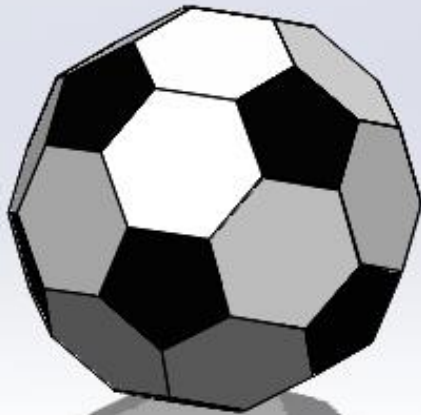
10/24/25
Friday
MAE 1001

Solidworks Crash Course 1! Your Gateway to 3D Creativity!



Session Outcomes

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Create 1/70th of a Soccer Ball!

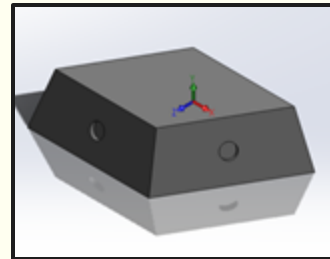
Today, we're going to introduce:

- **2D Sketching** a Part
- Converting 2D Sketches to a **3D Part**
- The Solidworks **Drawing** feature

These three skills will help you start your **professional career**.

At the end of next week's session, you will have a **customized** piece of this soccer ball!

Your Part!



Classrooms To Access Solidworks: Remotely and Physically

Janssen 2025

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SEH 3040

SEH 4040

TOMP 405

TOMP 406

Classes where you'll see Solidworks

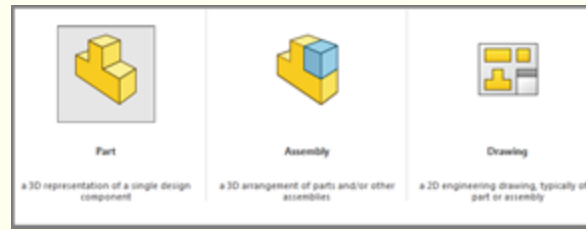
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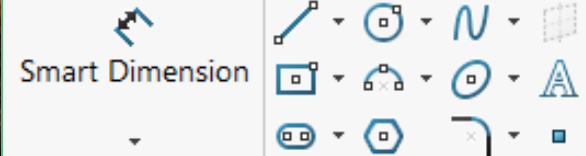
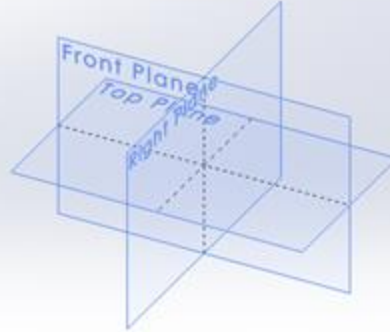
MAE 1001	Introduction to Mechanical and Aerospace Engineering	
MAE 1004	Engineering Drawing and Computer Graphics	
MAE 3187	Heat Transfer	
MAE 3191	Mechanical Design of Machine Elements	
MAE 3192	Manufacturing Processes and Systems	
MAE 3193	Mechanical Systems Design	
MAE 4151	Capstone	And many more... (as well as clubs, research, internships, etc.)

Part 1: 2D Sketches

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Sketch



1. Start a Sketch

This is the first step when modelling any part. It creates a 2D drawing that, using the wide variety of tools given by solidworks, can be made into any complex 3D shape.

2. Select a Plane

Orienting yourself in 3D space is crucial in solidworks, especially as you work with more and more complex models and assemblies, make sure that you know what plane you're working on.

3. Start with the basics!

Solidworks offers a variety of 2D modelling tools to help you get starting in your modelling journey. Play with these and get familiar with them as they will allow you to make more complex parts.

Let's Make our Part!

Hint: **Follow** along here and feel free **ask questions**, when we're done with this part you'll be **75% done** with your **CAD assignment!**

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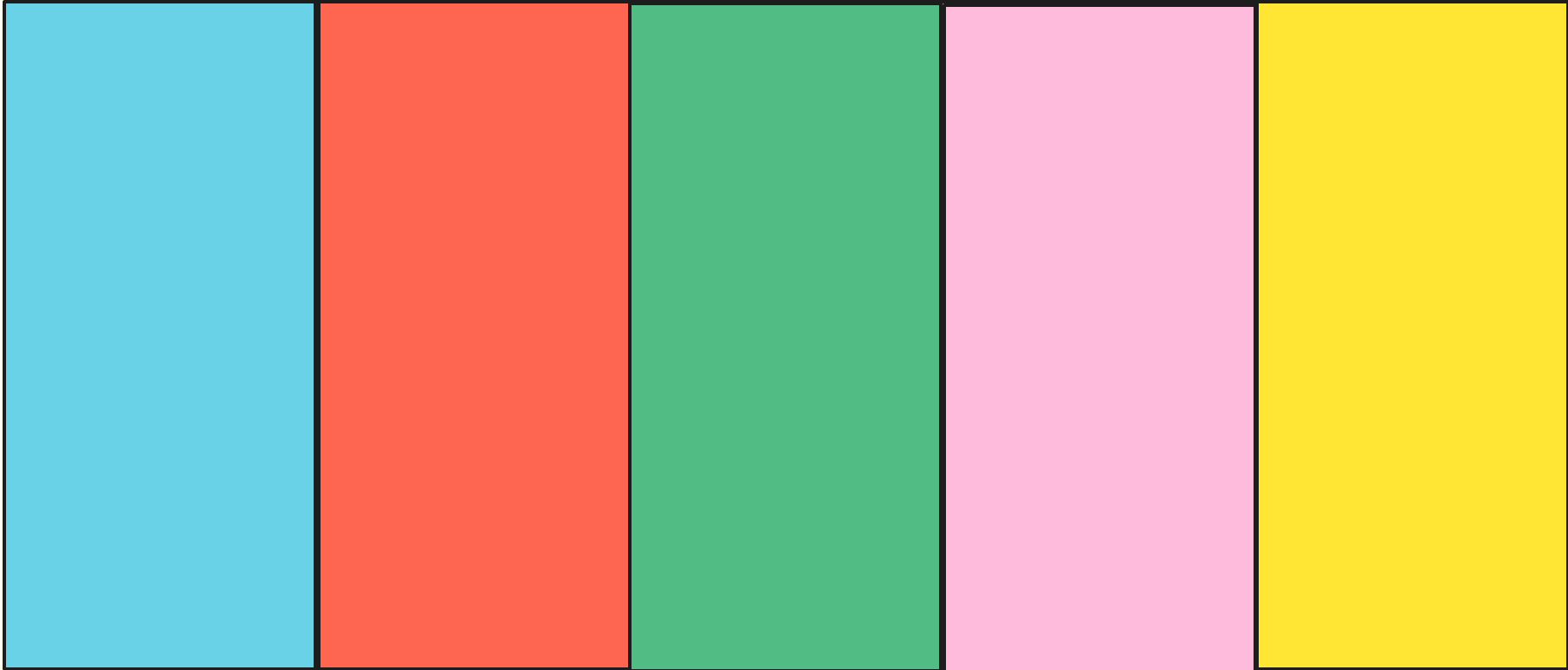
1.

2.

3.

4.

5.



Making our Part into a Drawing!

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First,

From the part, find the  (New) dropdown and select the Drawing option

Then,

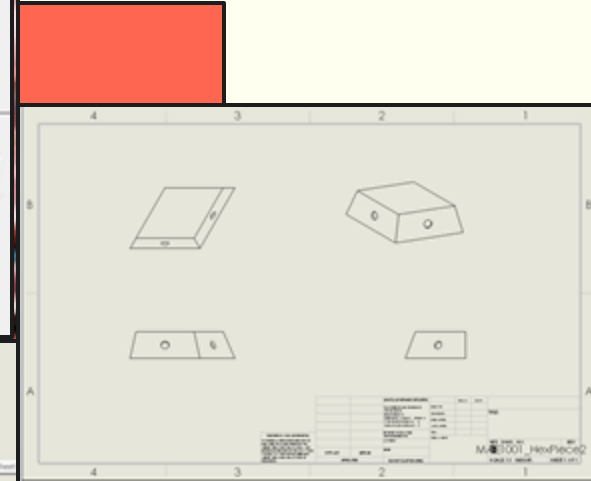
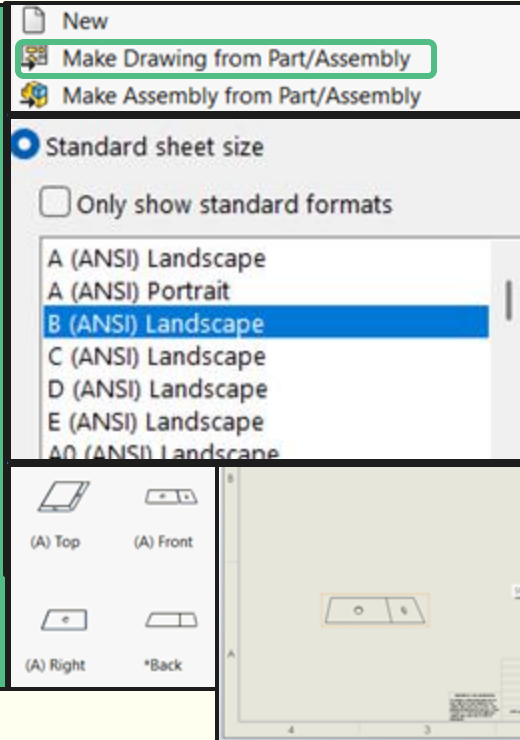
Turn off the standard formats option and find ANSI B (a common format)

Next,

Select the front face of the part and drag it to the bottom left of the drawing

Finally,

Solidworks *automagically* populates where you move the part to the side that should be viewed, click around the front face to finish your drawing.



Customization Criteria

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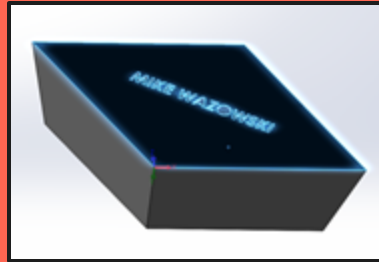
No Extrusions over 2cm!

Let's try to conserve our materials the best we can and not be wasteful while still having fun!



Your name must be on it!

Use the  (Text) tool to add your name to the drawing and either cut or extrude it out ~0.2 cm



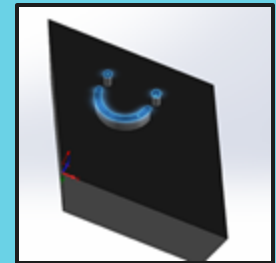
No Overhangs!

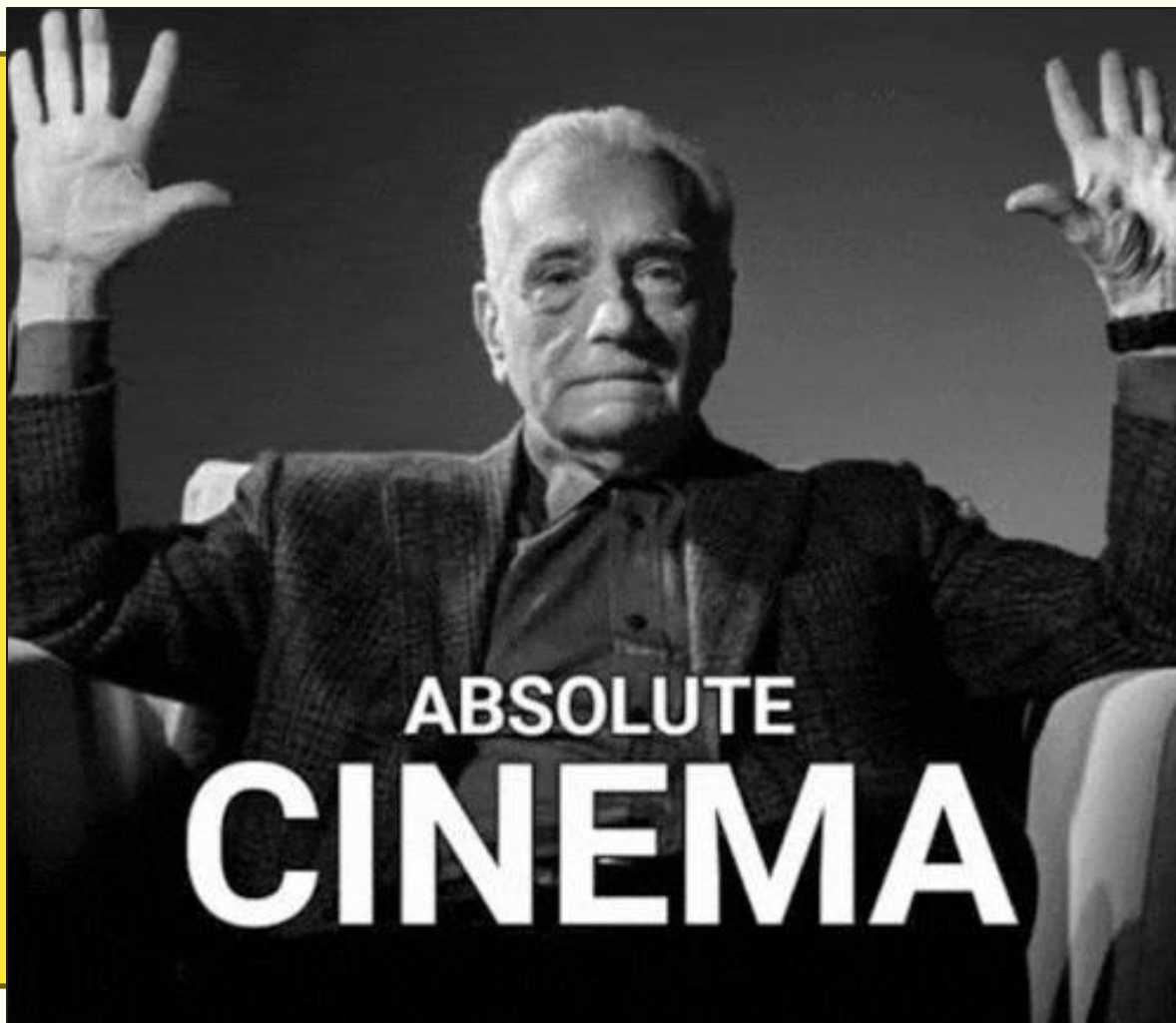
We are trying to print with **NO SUPPORTS**, we can do this as long as we have no overhangs on our prints



Have fun and be original!

Remember, this is supposed to be fun! Try something new, if it doesn't work, ask me, or a friend, or the internet.





Q&A

?

